

General Knowledge

Indian Geography

STUDY NOTES

Physiographic Divisions of India - Plateaus and Deserts



PHYSIOGRAPHIC DIVISIONS OF INDIA -

PLATEAUS AND DESERTS

The Peninsular Plateau

The Peninsular Plateau is the oldest and largest physiographic division in India, with elevations between 600 to 900 meters. Shaped like an irregular triangle, it is bordered by the Delhi Ridge (an extension of the Aravallis) in the northwest, the Raj Mahal hills in the east, the Gir range in the west, and the Cardamom hills in the south.

Formed by the drifting and breaking of the Gondwana landmass, the plateau comprises ancient crystalline, igneous, and metamorphic rocks. The region has undergone repeated uplifts, submergence, faulting, and fracturing, leading to diverse relief features. The northwest part has ravines and gorges, including the ravines of Chambal, Bhind, and Morena. Notably, the Deccan Trap in this plateau is a black soil area formed from volcanic igneous rocks.

The Peninsular Plateau can be divided into three main regions based on relief features:

- **The Central Highlands:** This area lies north of the Narmada River, covering most of the Malwa Plateau. The Aravallis lie to the north, and the Vindhyan Range to the south. Eastward, the plateau narrows and is known as Bundelkhand, Baghelkhand, and the mineral-rich Chotanagpur Plateau.
- **The Deccan Plateau:** South of the Narmada, this triangular plateau has the Satpura Range at its base and the Mahadev, Kaimur, and Maikal ranges as eastward extensions. Sloping gently eastward, it extends to the northeast as the Meghalaya, Karbi-Anglong Plateau, and North Cachar Hills.
- **The Northeastern Plateau:** An extension of the Peninsular Plateau, the Meghalaya Plateau is separated from the main plateau by a fault and consists of the Garo, Khasi, and Jaintia Hills, rich in minerals and experiencing heavy erosion due to monsoon rainfall.



Central Highlands (Madhya Bharat Plateau)

- **Location:** Spans across parts of **Madhya Pradesh, Rajasthan, Uttar Pradesh**, and parts of **Maharashtra**.
- **Geological Composition:** Mainly **sandstone, basalt**, and **sedimentary rocks**.
- **Height:** Elevation ranges between **300-800 meters** above sea level.
- **Main Rivers:** **Chambal River, Kali Sindh River, Banas River, Parbati River**, and **Parwan River**.

- **Topography:** The Central Highlands are marked by **rolling plateaus, ridges, and hills**. It includes the **Chambal Ravines**, known for their deep erosional valleys and badlands.

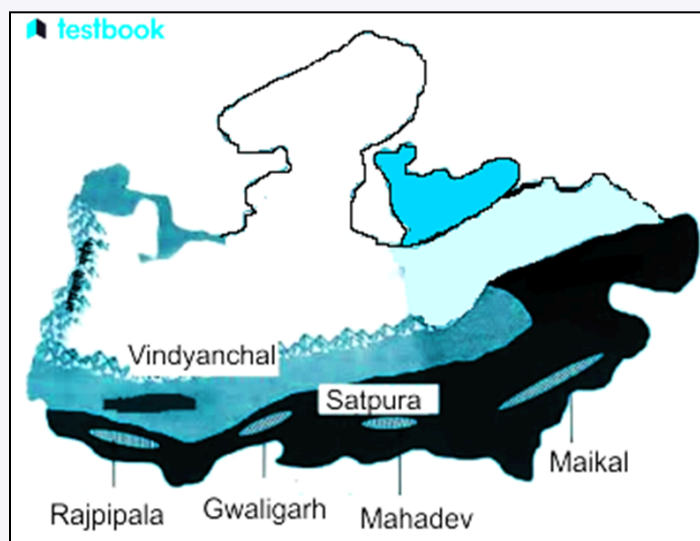
Subdivisions:

- **Bundelkhand Plateau:**
 - **Location:** Between **Yamuna River** in the north and **Vindhyan Range** in the south.
 - **Height:** Elevation varies between **300 to 600 meters**.
 - **Rivers:** The **Yamuna, Betwa, and Ken rivers** flow through this region.
 - **Other Features:** Known for **ridges, hills, and ravines** formed by river erosion.
- **Malwa Plateau:**
 - **Location:** Northwestern part of the Central Highlands, covering parts of **Madhya Pradesh**.
 - **Height:** Ranges from **500 to 700 meters**.
 - **Rivers:** The **Narmada, Mahi, and Chambal rivers**.
 - **Other Features:** The plateau is characterized by **flat-topped hills, basaltic lava deposits, and black soil**.
- **Chambal Valley (Ravines):**
 - **Location:** Between the **Madhya Bharat Plateau** and the **Bundelkhand Plateau**.
 - **Height:** Around **300 meters** above sea level.
 - **Rivers:** **Chambal River** forms the main feature of the valley.
 - **Other Features:** Known for **badlands** and **erosional features** formed by the Chambal River.

Vindychal Range and the Satpura Range:

Feature	Vindychal Range	Satpura Range
Location	Central India: Madhya Pradesh, Uttar Pradesh, Bihar	Central India: Madhya Pradesh, Maharashtra
Extent	1,000 km, east-west	800 km, east-west
Highest Peak	Dun (752 m)	Dhupgarh (1,350 m)

Geological Formation	Sandstone, Archaean rocks	Granite, basalt, sandstone
Major Rivers	Narmada, Son, Ken	Narmada, Tapi, Mahi
Climate	Semi-arid to temperate	Tropical, with monsoon rains
Vegetation	Scrub forests, dry deciduous	Dense forests, rich biodiversity
Ecological Significance	The natural barrier between Northern and Southern India	Rich in wildlife, national parks, sanctuaries



Deccan Plateau

- **Location:** Covers a large part of southern India, extending across **Maharashtra, Karnataka, Telangana, Andhra Pradesh, Tamil Nadu**, and parts of **Madhya Pradesh**.
- **Geological Composition:** Primarily made of **basaltic lava** (Deccan Traps) formed by ancient volcanic activity.
- **Height:** Elevation ranges from **500 meters** in the north to over **1,000 meters** in the southern parts.
- **Main Rivers:** **Godavari, Krishna, Kaveri, Mahanadi**, and **Tungabhadra**.
- **Topography:** The plateau is characterized by **flat-topped hills, river valleys**, and **basaltic rock formations**.

Subdivisions:

- **Maharashtra Plateau:**
 - **Location:** Covers most of **Maharashtra**.
 - **Height:** Ranges between **600 to 1,000 meters**.
 - **Rivers:** The **Godavari**, **Krishna**, and **Tungabhadra** rivers flow through this region.
 - **Other Features:** Known for **black soil** and **lava plateau**. It is rich in **minerals**, especially **bauxite**, **iron ore**, and **coal**.
- **Karnataka Plateau:**
 - **Location:** Covers most of **Karnataka**.
 - **Height:** Ranges between **600 to 900 meters**.
 - **Rivers:** The **Krishna**, **Tungabhadra**, and **Kaveri** rivers.
 - **Other Features:** The region is marked by **hills**, **forests**, and **fertile soils** suitable for agriculture. The **Western Ghats** lie on the western edge, contributing to the plateau's high biodiversity.
- **Telangana Plateau:**
 - **Location:** Covers the state of **Telangana**.
 - **Height:** Varies from **500 to 1,000 meters**.
 - **Rivers:** The **Godavari** and **Krishna** rivers.
 - **Other Features:** Known for **granite** and **limestone** deposits. It also has **rich deposits of coal** and **limestone**.
- **Andhra Plateau:**
 - **Location:** Covers the state of **Andhra Pradesh**.
 - **Height:** Around **500 to 900 meters**.
 - **Rivers:** **Godavari** and **Krishna**.
 - **Other Features:** Famous for its **basaltic terrain**, **bauxite deposits**, and **forests**.
- **Tamil Nadu Plateau:**
 - **Location:** Spreads across southern **Tamil Nadu**.
 - **Height:** Ranges from **300 to 600 meters**.
 - **Rivers:** **Kaveri** and its tributaries.
 - **Other Features:** The plateau is known for its **agriculture**, **granite hills**, and **rich soil** for crops like **rice** and **cotton**.

Northeastern Plateau

- **Location:** Lies in the **northeastern part of India**, covering **Arunachal Pradesh, Assam, Nagaland, Manipur, Mizoram, Meghalaya**, and parts of **Tripura**.
- **Geological Composition:** Made up of **sandstone, limestone, and schist**.
- **Height:** Varies between **300 to 1,500 meters**.
- **Main Rivers:** **Brahmaputra, Barak, and Irawaddy**.
- **Topography:** The terrain is marked by **hills, valleys, and high plateaus**. This region is known for its **dense forests** and **high rainfall**.

Subdivisions:

- **Meghalaya Plateau:**
 - **Location:** Covers **Meghalaya, Assam**, and parts of **Nagaland**.
 - **Height:** Between **900 to 1,500 meters**.
 - **Rivers:** The **Brahmaputra, Barak, and Irawaddy** rivers.
 - **Other Features:** Known for its **limestone** caves and **biodiversity**.
- **Karbi Anglong Plateau:**
 - **Location:** Spans across parts of **Assam**.
 - **Height:** Ranges between **600 to 900 meters**.
 - **Rivers:** The **Brahmaputra** and its tributaries.
 - **Other Features:** The plateau is covered with **dense forests** and is rich in **coal** and **limestone**.
- **Naga Hills Plateau:**
 - **Location:** Spans parts of **Nagaland** and **Myanmar**.
 - **Height:** Between **800 to 2,000 meters**.
 - **Rivers:** **Chindwin River** and **Brahmaputra** tributaries.
 - **Other Features:** Known for its **hilly terrain, rich biodiversity, and high rainfall**.

Plateau Type	Subdivisions	Location	Elevation	Rivers	Key Features
Central Highlands	Bundelkhand, Malwa, Chambal	Madhya Pradesh, Rajasthan, Uttar Pradesh	300-800 meters	Chambal, Kali Sindh, Banas, Parbati	Rolling hills, ravines, badlands

Deccan Plateau	Maharashtra, Karnataka, Telangana, Andhra, Tamil Nadu	Maharashtra, Karnataka, Telangana, Andhra Pradesh, Tamil Nadu	500-1,500 meters	Godavari, Krishna, Kaveri, Mahanadi	Lava plateaus, basaltic terrain, minerals
Northeastern Plateau	Meghalaya, Karbi Anglong, Naga Hills	Arunachal Pradesh, Assam, Nagaland, Mizoram, Meghalaya	300-2,000 meters	Brahmaputra, Barak, Irawaddy	Hilly terrain, limestone caves, forests

Western Ghats (Sahyadris)

The **Western Ghats** are continuous mountain ranges that run parallel to the western coast of India, forming a major barrier to the monsoon winds and contributing significantly to India's climate and biodiversity.

Formation and Structure:

- **Formation:** Formed by the **subduction of the Arabian Plate** and the tilting of the Indian Peninsula towards the east during the **Himalayan uplift**.
- The range is characterized by **block mountains** and **escarpments**.
- The western slope is steep, forming **Treppen** or step-like escarpments, while the eastern slope is more gradual, merging into the plateau.

Geographical Spread:

- Stretches from **Tapi Valley** in the north to **Kanyakumari** in the south.
- It is spread across six states: **Gujarat, Maharashtra, Goa, Karnataka, Tamil Nadu, and Kerala**.
- It is one of the **hottest biodiversity hotspots** in the world and a **UNESCO World Heritage site**.

- The Western Ghats are known by different local names such as **Sahyadri** in **Maharashtra**, **Nilgiri hills** in **Karnataka** and **Tamil Nadu** and **Anaimalai hills**, and **Cardamom hills** in **Kerala**.
- The height of the Western Ghats progressively increases from north to south.
- The highest peaks include the **Anai Mudi (2,695 meters)** and the **Doda Betta (2,637 meters)**.

Sections of the Western Ghats:

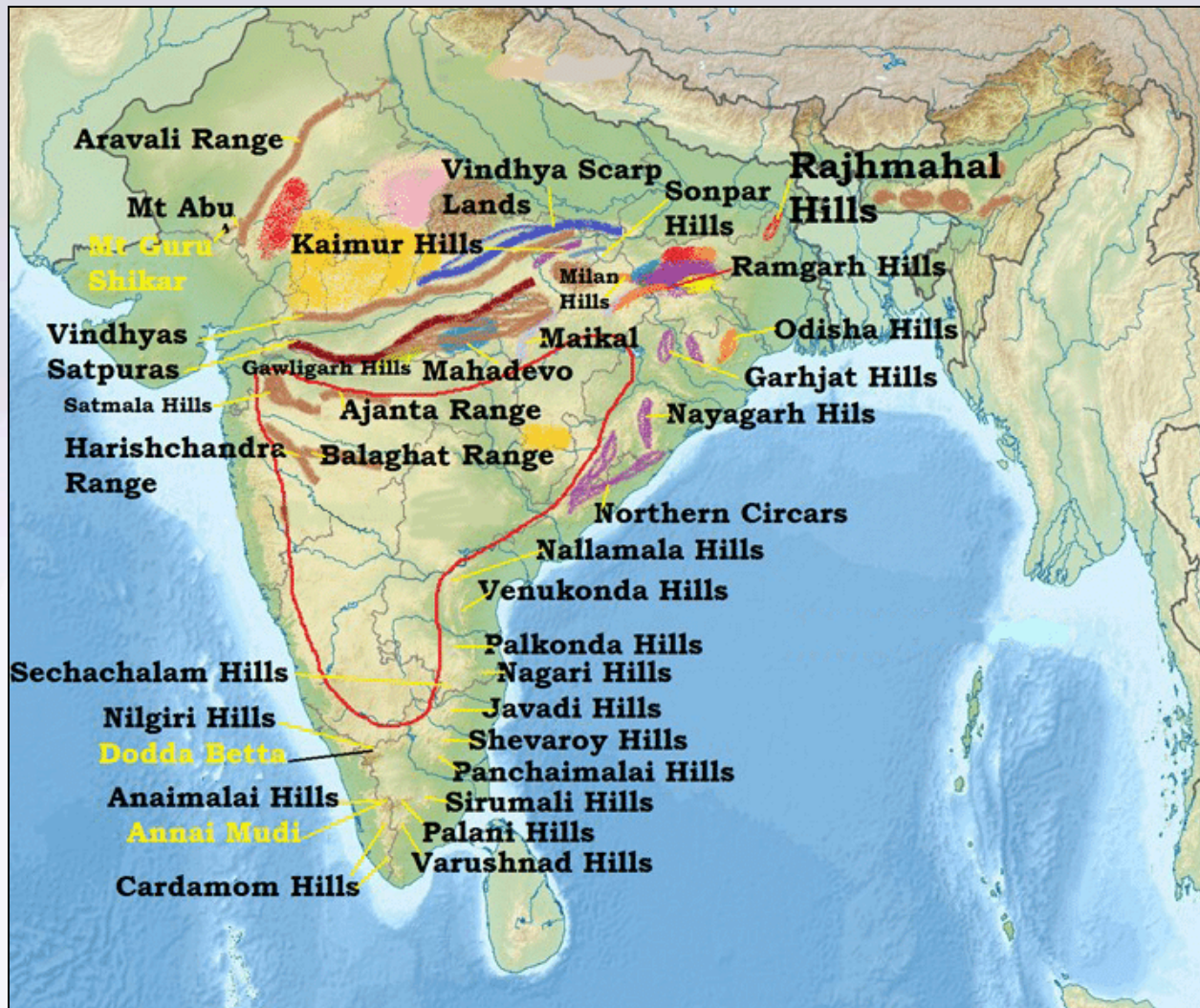
- **Northern Western Ghats (From Tapi Valley to 16°N):**
 - Composed mainly of **basaltic lava** from the **Deccan Traps**.
 - Average elevation: **1,200 meters**.
 - The **highest peak** is **Kalsubai** (1,646 meters).
 - Other notable peaks: are **Salher** (1,567 meters), **Mahabaleshwar** (1,438 meters), and **Harishchandragarh** (1,424 meters).
 - This region is **rugged** and dissected by numerous rivers like the **Godavari**, **Krishna**, and **Tungabhadra**.
 - Major river valleys: **Tungabhadra**, **Mahi**, **Narmada**, and **Tapi**.
- **Central Western Ghats (Around 16°N to 13°N):**
 - Composed of **granite** and **gneiss**, which are older and more **weathered** than the northern basalt.
 - More continuous in nature and has **dense forests**, making it ecologically rich.
 - **Idukki Hills** (Kerala), **Anamudi** (2,695 meters) – the highest peak in the Western Ghats, located in the **Eravikulam National Park**.
 - Rivers like the **Periyar**, **Kaveri**, and **Bhavani** originate in this region.
- **Southern Western Ghats (South of 13°N):**
 - Characterized by **higher peaks** and **wider valleys**.
 - Includes **Nilgiri Hills** and the **Anamalai Hills**.
 - **Highest peaks:** **Meghamalai** (1,590 meters), **Agasthyakoodam** (1,868 meters), **Anamudi** (2,695 meters).
 - The region has significant biodiversity, especially in the **Silent Valley** and **Parambikulam**.
 - Rivers: **Kaveri**, **Bhavani**, **Periyar**, and **Narmada**.

Ecological Significance:

- The Western Ghats are a **UNESCO World Heritage Site** and a **biodiversity hotspot** with a large number of endemic species.
- Home to **tropical rainforests, savannas, and wetlands**.
- Influences monsoon patterns by intercepting the southwest monsoon winds and causing heavy rainfall, especially in Kerala and coastal regions.

Important hills of the Western Ghats from North to South

Hill Range	Location	Additional Information
Satpura Range	Madhya Pradesh, Maharashtra, Gujarat	Forms the northern edge of the Western Ghats.
Satmala Hills	Maharashtra	Part of the Deccan Plateau extends to the Thane district.
Harishchandra Range	Maharashtra	Known for Harishchandragad Fort, steep terrain.
Baba Budan Hills	Karnataka	Famous for the Baba Budangiri range, sacred to Hindus and Muslims.
Nilgiri Hills	Tamil Nadu, Kerala, Karnataka	Known for its biodiversity, tea gardens, and the Nilgiri Biosphere Reserve.
Anaimalai Hills	Tamil Nadu, Kerala	Known as the Elephant Mountains; part of the Western Ghats.
Anai Mudi	Kerala	The highest peak in the Western Ghats is located in the Anaimalai Hills.
Agasthyamalai	Kerala, Tamil Nadu	Named after the sage Agastya, a major biodiversity hotspot.
Cardamom Hills	Kerala, Tamil Nadu	Known for its cardamom plantations, the southernmost range of India.



Economic and Cultural Importance:

- Known for **tea**, **coffee**, and **spice plantations**, especially in **Kerala**, **Karnataka**, and **Tamil Nadu**.
- **Tourism**: Popular for **hill stations** like **Mahabaleshwar**, **Munnar**, and **Coorg**.
- **Agriculture**: The region supports agriculture due to its fertile soils, including **rice**, **spices**, and **tropical fruits**.

Eastern Ghats

The **Eastern Ghats** are discontinuous mountain ranges that run parallel to the eastern coast of India, from the **West Bengal** coast in the north to the **Tamil Nadu** coast in the south. **Mahendragiri (1,501 meters)** is the highest peak in the Eastern Ghats.

Formation and Structure:

- The Eastern Ghats are older and more **dissected** compared to the Western Ghats.
- They are formed from **crystalline rocks**, including **gneisses** and **granites**, and are **weathered** over time.
- The **Eastern Ghats** are not as continuous as the Western Ghats and are marked by gaps and valleys through which rivers flow.

Geographical Spread:

- Runs parallel to the **eastern coast** of India, from **Odisha** and **West Bengal** in the north to **Tamil Nadu** in the south.
- Extends over **Odisha, Andhra Pradesh, Telangana, Karnataka, and Tamil Nadu**.

Sections of the Eastern Ghats:

- **Northern Eastern Ghats (Odisha and Andhra Pradesh):**
 - Extends from **West Bengal** to **Andhra Pradesh**.
 - Includes **Similipal Hills** and **Kailashgiri**.
 - The **highest peak** is **Mahendragiri** (1,501 meters) in **Odisha**.
 - Rivers: **Mahanadi, Godavari, Krishna, and Vamsadhara**.
- **Central Eastern Ghats (Andhra Pradesh and Telangana):**
 - The **Ananthagiri** and **Nallamala Hills** are prominent here.
 - Known for **forests** and rich biodiversity.
 - Major rivers: **Krishna** and **Godavari**.
- **Southern Eastern Ghats (Tamil Nadu):**
 - The range becomes lower and more **dissected** in the southern part.
 - Prominent hills include **Nilgiri Hills, Anaimalai Hills, and Shevaroy Hills**.
 - **Highest Peak: Javadi Hills**.
 - The **Palar** and **Vaigai** rivers originate from this section.

Ecological Significance:

- The Eastern Ghats are not as biologically rich as the Western Ghats, but they still contain several **national parks** and **wildlife sanctuaries**, such as the **Similipal Wildlife Sanctuary** (Odisha) and **Nallamala Forests** (Andhra Pradesh).

- Important for the **flora and fauna** of the region, including **tiger reserves** and habitats for **elephants** and **deer**.

Economic Importance:

- The Eastern Ghats are home to **rich mineral resources**, particularly **bauxite** and **iron ore**, with **mining** industries in states like **Odisha** and **Andhra Pradesh**.
- **Agriculture**: The fertile valleys support **rice**, **cotton**, and **tobacco** crops.

Important hills of the Eastern Ghats from North to South

Hill Range	Location	Additional Information
Mahendragiri Hills	Odisha, Andhra Pradesh	One of the highest peaks in the Eastern Ghats.
Simhachalam Hills	Andhra Pradesh	Known for the Simhachalam Temple, a significant religious site.
Kailasagiri Hills	Andhra Pradesh	Famous for scenic views and tourist destinations near Visakhapatnam.
Nallamala Hills	Andhra Pradesh, Telangana	Rich in biodiversity, home to the Nallamala forest.
Palkonda Hills	Andhra Pradesh	Located in the Rayalaseema region, characterized by steep ridges.
Rayadurg Hills	Andhra Pradesh	Located in the Rayadurg region of Andhra Pradesh, part of the Eastern Ghats.
Velikonda Hills	Andhra Pradesh, Tamil Nadu	Known for its granite ridges and ecological significance.
Javadi Hills	Tamil Nadu	Famous for the Javadi Hills plateau and rich flora.

Shevaroy Hills	Tamil Nadu	Known for the hill station Yercaud, popular for coffee plantations.
Anaimalai Hills	Tamil Nadu, Kerala	Part of the Southern Eastern Ghats, located in Tamil Nadu and Kerala.

Key Differences Between Western and Eastern Ghats:

Feature	Western Ghats	Eastern Ghats
Continuity	Continuous mountain range	Discontinuous and fragmented
Height	Generally higher (up to 2,695 m) Highest Peak - Anamudi	Lower in height (up to 1,501 m) Highest Peak - Mahendragiri
Rivers	Mainly flowing rivers (e.g., Godavari, Krishna, Kaveri)	Mainly flowing rivers (e.g., Mahanadi, Godavari, Krishna)
Ecological Significance	Rich in biodiversity, the UNESCO World Heritage	Moderate biodiversity, home to some reserves
Geological Composition	Basaltic lava, granites, gneisses	Crystalline rocks, gneisses, granites

The Thar Desert (Great Indian Desert)

The **Thar Desert** is one of the most prominent geographical features in the northwestern part of the Indian subcontinent. Known for its arid conditions, unique ecosystem, and historical significance, the desert forms a natural boundary between **India** and **Pakistan**.

- **Alternate Names:** Great Indian Desert, Marusthali (meaning 'Land of Deserts' in Sanskrit).
- **Area:** Spans about **200,000 square kilometers** in total, with approximately **85% of the desert** located within India (around **170,000 km²**), primarily in **Rajasthan**. It also extends into **Gujarat, Punjab, and Haryana** in India, and the **Sindh** province in Pakistan.

- **Global Rank:** The **17th-largest desert** in the world and the **9th-largest hot subtropical desert**.
- **Geographical Boundaries:**
 - To the **west**, it is bordered by the **Indus River Plain**.
 - To the **north and northeast**, it meets the **Punjab Plain**.
 - The **Aravalli Range** forms the **southeastern boundary**.
 - To the **south**, it is bordered by the **Rann of Kachchh**.



Geographical and Topographical Features:

- **Location:** Lies northwest of the **Aravalli Hills**.
- **Surface Features:** The surface consists of **undulating topography** shaped by **wind action** and **physical weathering**.
 - Features **longitudinal dunes**, **barchans** (crescent-shaped dunes), and **sandy plains**.
 - **Barchans** are crescent-shaped dunes formed by wind blowing in one direction.

- **Mushroom Rocks and Oases:** These are prominent natural formations, with oases found primarily in the southern region of the desert.
- **Shifting Sand Dunes:** The sand dunes in the Thar Desert shift frequently due to the constant wind action.

Climate:

- **Average Annual Rainfall:** Below **150 mm** per year, making it a **hyper-arid region**.
- **Summer:** Extremely hot, with temperatures often reaching up to **50°C**. Dust storms and strong winds are typical, especially between **May and June**, where wind speeds can reach **140-150 km/h**.
- **Winter:** In **January**, the desert experiences cold temperatures between **5°C to 10°C**, with occasional frost.
 - **Dust Storms:** These are common in the summer months, contributing to the extreme dryness and low visibility.

Geological Features:

- **Mesozoic Era:** The desert was believed to have been underwater during the **Mesozoic era**. Fossilized **wood** and **marine deposits** have been found in places like **Aakal** and **Brahmsar** near **Jaisalmer**, with some fossils estimated to be around **180 million years old**.
- **Akal Wood Fossil Park:** A **National Geological Monument of India**, located in **Jaisalmer**, it contains fossilized tree trunks and other ancient remains, providing evidence of the desert's once-aquatic past.

Rivers and Water Bodies:

- **Ephemeral Rivers:** Most of the rivers in the Thar Desert are **ephemeral**, meaning they only flow seasonally or after rainfall.
 - **Luni River:** The primary river in the region, it originates in the **Aravalli Range** and flows towards the south, but it eventually disappears into the sands, and its flow is often intermittent.
 - Other smaller rivers and streams often vanish into the **sand** and exhibit **inland drainage**, joining **lakes** or **playas** (dry lake beds).

- **Oases:** Found primarily in the southern part of the desert, oases are crucial for sustaining life and vegetation. Examples include **Phalodi**, **Jaisalmer**, and **Barmer**, where underground water sources support agriculture and human settlements.

Flora and Fauna:

- Despite its harsh conditions, the Thar Desert supports a variety of plant and animal life.

Flora:

- **Cacti**, **acacia**, and **sacrificial plants** like **khejri** (*Prosopis cineraria*) and **thar** (*Salvadora oleoides*) are common.
- **Desert shrubs** and **grass species** are adapted to survive in low-water conditions.

Fauna:

- **Mammals:** Includes **blackbuck antelope**, **chinkara** (Indian gazelle), **desert fox**, and **wild boar**.
- **Reptiles:** Common species include **sand boas** and **lizards**.
- **Birds:** The region is home to the **Indian bustard**, **Desert Eagle Owl**, and **sand grouse**.
- **Insects:** Includes **locusts**, **desert beetles**, and various species of **scorpions** and **termites**.

Human Settlement and Economic Activities:

- The **Thar Desert** is sparsely populated, but there are significant settlements in cities like **Jaisalmer**, **Barmer**, **Jodhpur**, and **Phalodi**.
- **Agriculture:** Limited irrigation from underground water sources supports the cultivation of **wheat**, **barley**, and **millets**. The desert also supports **livestock** farming, particularly **camels**, **sheep**, and **goats**.
- **Minerals:** The desert region has deposits of **salt**, **gypsum**, **limestone**, and **phosphates**. **Oil** reserves are also present, especially near **Barmer**.
- **Tourism:** The desert's unique landscape, historical forts, and palaces in places like **Jaisalmer** attract numerous tourists.

Ecological Importance:

- The **Western Ghats** and **Rann of Kachchh** ecosystems interact with the Thar Desert, creating unique biodiversity corridors.
- The Thar Desert plays a critical role in **climate regulation** and is an important part of India's **hydrological cycle**, despite its arid conditions.
- As one of the **world's biodiversity hotspots**, it contributes significantly to global ecological health.